CARMA

Multi expert decision engine, NSE equities | Out of sample results, 2023 to 2024

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Test Setup

Universe	26 NSE equities, large and mid cap. 23 produced trades; MARUTI, ULTRACEMCO and DIXON produced none under any configuration.
Training window	2018 to 2022. Technical expert 2019 to 2022, pattern expert 2018 to 2023 (5 fold CV), meta model 2019 to 2022.
Test window	2023-03-16 to 2024-12-19. 1.76 years. Never used in training or selection.
Direction	Long only.
Costs	0.05 percent slippage per side plus Rs 20 brokerage per order, charged on notional position value.
Sizing	1R per trade, where 1R is 1 percent of that stock's running equity, compounding from Rs 100,000. Meta model confidence acts as an entry gate at 0.75, not as a size multiplier (mean confidence of taken trades 0.85).
Exits	Structural stop from ATR geometry, three stage trailing stop, 25 bar timeout.

Results

	Gross	Net, 1x costs	Net, 2x costs
Trades	113	113	113
Win rate	62.8%	**61.9%** (70/113)	60.2% (68/113)
Expected R per trade	+0.453R	**+0.385R**	+0.317R
Stocks profitable	19 / 23	19 / 23	19 / 23

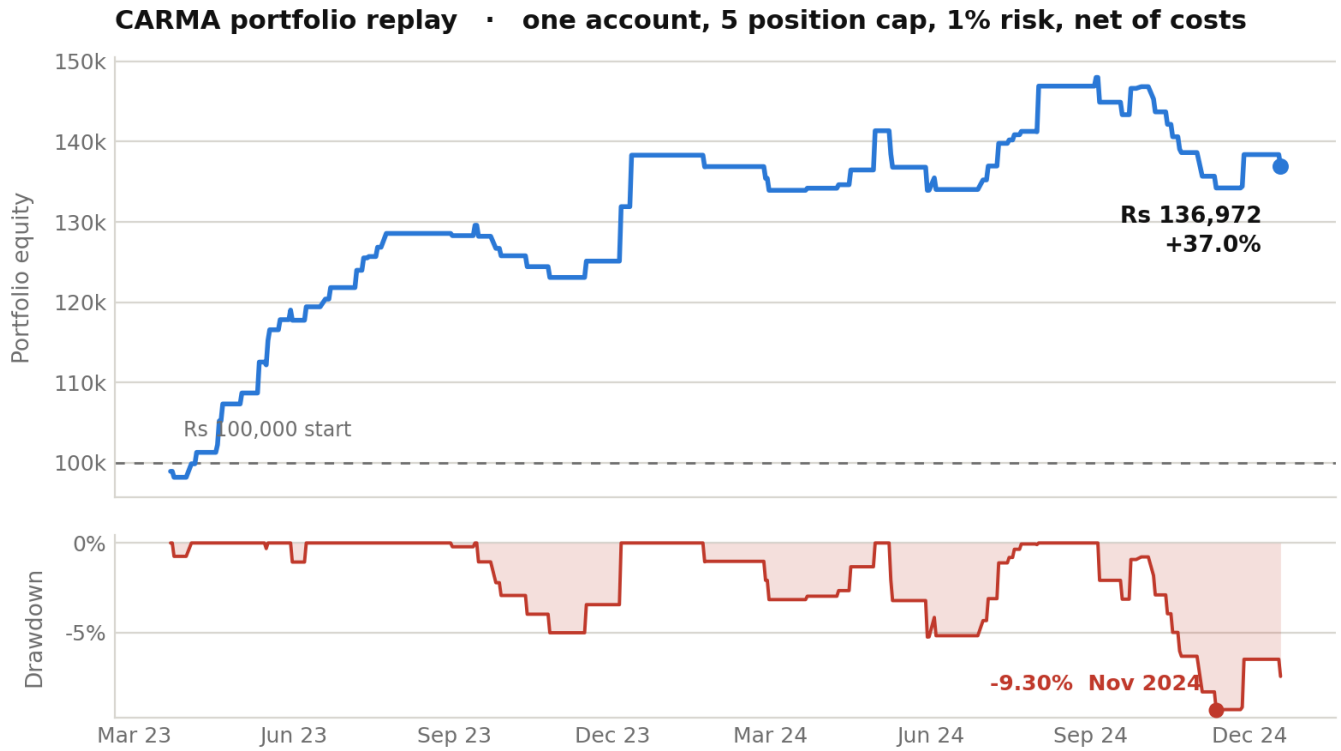
Does the full stack beat the best single expert? Yes. The Technical Volume expert alone produces +0.204R gross across 494 trades at a 45.7 percent win rate. The full pipeline produces +0.385R net across 113 trades at 61.9 percent. The stack trades roughly one quarter as often and earns close to double the expected R per trade, net of costs.

Transaction Costs

Cost per trade	Mean 0.068R, median 0.061R, range 0.048R to 0.224R.
Total cost, 113 trades	Rs 7,803. Brokerage Rs 4,520, slippage Rs 3,283.
Sensitivity	At 2x costs the edge falls from +0.385R to +0.317R and the strategy stays profitable on the same 19 of 23 stocks. Low turnover is why: 113 trades over 1.76 years, averaging 13.3 bars held.

Portfolio Simulation

The backtest simulates each stock against its own Rs 100,000, which is not how the strategy would be traded. The 113 trades are replayed below through a single shared account: risk 1 percent of current portfolio equity per trade, at most 5 concurrent positions, compounding on net R. Signals arriving when the book is full are skipped. 86 of 113 signals were taken; the 27 skipped averaged +0.351R against +0.396R for those taken, so the constraint is not selecting the better trades.



Total return	+36.97 percent over 1.75 years, Rs 100,000 to Rs 136,972.
Annualised return	**+19.73 percent**.
Sharpe ratio	**1.28** on monthly returns, risk free rate 0. The daily figure is 1.75, but equity realises at trade exit rather than marking to market, which suppresses day to day variation and flatters it. The monthly number is the one to use.
Maximum drawdown	**9.30 percent**, trough 12 November 2024. Longest stretch below a prior peak 75 business days.
Monthly profile	11 of 21 months positive. Best +10.54 percent, worst −5.59 percent, mean +1.70 percent, standard deviation 4.61 percent.
Exposure	At least one position open on 86.3 percent of business days. Mean 2.66 concurrent positions, at the 5 position cap on 21 percent of days.
Holding period	13.3 bars mean, median 12, range 1 to 25.

Sensitivity. The headline figure is one point on three axes, so all three are given.

	CAGR	Sharpe	Max DD
Transaction costs, at cap 5 and 1 percent risk			
Gross, no costs	+23.63%	1.46	−8.41%
1x costs (headline)	+19.73%	1.28	−9.30%
2x costs	+15.95%	1.08	−10.18%
Concurrent position cap, at 1 percent risk			
2 positions (46 taken)	+4.55%	0.75	−4.58%
3 positions (63 taken)	+7.88%	0.85	−6.66%
5 positions (86 taken)	+19.73%	1.28	−9.30%
8 positions (107 taken)	+23.09%	1.25	−10.28%
Uncapped (113 taken)	+25.00%	1.26	−10.28%
Risk per trade, at cap 5			
0.5 percent	+9.81%	1.27	−4.71%
1.0 percent (headline)	+19.73%	1.28	−9.30%
2.0 percent	+39.64%	1.28	−18.11%
3.0 percent	+59.17%	1.28	−26.42%

All Sharpe figures above are monthly, consistent with the headline. Sharpe is flat across risk levels, which is the expected behaviour and a check that the accounting is sound. The position cap behaves differently on a risk adjusted basis than on return: raising it from 5 to uncapped adds roughly 5 percentage points of annual return but does not improve Sharpe (1.28 against 1.26) and deepens the drawdown, so the capped book is not giving anything up. Tightening below 5 does hurt on both counts. On a 21 month sample these differences are indicative rather than significant.

Methodology Correction

These figures differ from an earlier write up of this system, which reported 148 trades at a 67.2 percent win rate and +0.669R. Two defects were found in the backtest on 2 October 2026 and corrected before producing this sheet.

Lookahead in pattern features. Swing point detection used a window centred on the current bar, extending three bars forward. Nine pattern features derived from it, and those features fed the entry gate. A point in time reconstruction test confirmed the leak: on a sampled stock, 74 of 145 bars had at least one feature change when later bars were withheld. The fix applies a three bar confirmation lag, so a swing point at bar t becomes readable only at bar $t+3$. Detection logic is unchanged. After the fix the same test shows 0 of 145 bars affected. Cost of the fix: win rate 65.4 to 63.2 percent, expected R +0.626 to +0.469 gross.

Transaction costs were documented but never implemented. The earlier write up described the cost model above; no such code existed. It is now applied per trade on notional value. Cost of the fix: +0.469R to +0.385R.

Known Limitations

1. **The portfolio figures are a replay, not a re-run.** The 113 trades were selected by a backtest in which the meta model gate saw each stock's own drawdown. In a true portfolio system it would see portfolio drawdown and become more selective across every name at once, so the trade set would differ. The replay gives the right order of magnitude; rebuilding the backtest loop to iterate by date across all stocks with shared state is outstanding.
2. **The pattern model was itself trained on unlagged features.** The lag fix was applied to the backtest only. The model therefore receives inputs whose distribution differs from its training data, so these results are a lower bound on the lag corrected pipeline rather than its true level. Retraining the pattern expert on lagged features is outstanding.
3. **Out of universe generalisation not yet re-tested.** An earlier run on 74 stocks outside this universe produced 60.9 percent and +0.430R, but it was produced before the lookahead fix and has not been repeated since.
4. **Partial sentiment contamination.** On a cache miss the sentiment module fetches current headlines. 30 of the 113 trades hit a miss and were scored against 2026 news rather than contemporaneous news.
5. **Thin per stock samples.** Several names carry only 1 to 3 trades. The universe aggregate of 113 is the meaningful figure; the per stock table shows dispersion, not per name significance.

Per Stock Results, Net of 1x Costs

Stock	Trades	Win rate	Gross R	Net R	Max DD
ICICIBANK	5	80.0%	+1.327	+1.268	1.06%
TITAN	2	100.0%	+1.413	+1.350	0.00%
RELIANCE	5	60.0%	+1.027	+0.956	1.08%
NTPC	6	66.7%	+0.966	+0.905	1.35%
IRCTC	6	66.7%	+0.937	+0.880	2.10%
INFY	2	100.0%	+0.901	+0.837	0.00%
PERSISTENT	2	100.0%	+0.760	+0.679	0.00%
ASIANPAINT	7	85.7%	+0.727	+0.654	1.06%
AXISBANK	7	85.7%	+0.696	+0.631	1.07%
HDFCBANK	3	33.3%	+0.504	+0.434	2.13%
POWERGRID	4	50.0%	+0.440	+0.376	1.29%
TCS	4	50.0%	+0.401	+0.343	0.83%
LT	2	50.0%	+0.360	+0.304	0.35%
KOTAKBANK	5	40.0%	+0.301	+0.229	3.20%
COALINDIA	11	54.5%	+0.247	+0.179	2.09%
BAJFINANCE	4	50.0%	+0.199	+0.128	2.11%
WIPRO	11	63.6%	+0.198	+0.121	2.15%
NESTLEIND	4	75.0%	+0.178	+0.121	1.06%
ONGC	8	62.5%	+0.170	+0.115	2.09%
SBIN	6	50.0%	−0.030	−0.099	2.11%
SUNPHARMA	3	33.3%	−0.173	−0.278	1.18%
HINDUNILVR	5	40.0%	−0.260	−0.319	2.11%
APOLLOHOSP	1	0.0%	−1.000	−1.183	1.18%
Universe	**113**	**61.9%**	**+0.453**	**+0.385**	**3.20%**

On the four unprofitable names. They are reported rather than excluded. Removing them would mean selecting the universe using out of sample results, which would invalidate the out of sample claim entirely. APOLLOHOSP carries a single trade and is noise rather than evidence. HINDUNILVR was described in the earlier write up as resolved by the derivatives filter; that result did not survive the correction.

Trade Log

The full log accompanies this sheet as `trade_log_final.csv`, 113 rows, one per trade: symbol, entry and exit date and price, exit reason, bars held, gross R, cost in rupees and in R, net R, net R at 2x, risk amount, running equity, and the three expert probabilities at entry. Two further logs are available on request, the pre correction run (127 trades, 65.4 percent, +0.626R) and the lag corrected run before costs (117 trades, 63.2 percent, +0.469R), which together isolate each correction.